CARPENTER AND JOINER

1. **PRELIMINARY AND GENERAL:**
Read and note all clauses of the Preliminary and General which apply to all works of this section.

2. **EXTENT OF WORK:**
The work of this section shall include all labour, materials, equipment necessary to carry out and complete the carpentry as shown, or as further required by this specification. Together with any other items of work reasonably inferred as part of this section.

3. **ATTENDANCE AND PROTECTION:**
Attend upon all other trades providing all blockings, fixings, trims, nogging as necessary for the full completion of their respective works and make good after all trades.

4. **MATERIALS AND WORKMANSHIP:**
All material used shall be the best of their respective class and type specified. Any materials which in the opinion of the Owner, Loan Body or Territorial Authority are not up to standard, to be removed immediately from the site. All work shall be carried out in a workmanlike manner in accordance with best trade practice and as or where especially specified. Any work that is intended or implied but not specifically shown, mentioned or specified as necessary for the proper completion of the building shall be included. All work shall comply with the relevant NZS requirements, particularly NZS 3604.

5. **TIMBER:**
All timber used shall conform to NZS 3602 requirements, "Code of practice for specifying time and wood-based products for use in building". Timbers shown in the Schedule of Timbers to be treated with an approved preservative process in plants licensed by the Timber Preservation Authority.

6. **DAMP PROOFING:**
All timber to be protected from dampness with 3-ply bituminous felt or other approved damp proofing material when in contact with concrete or brickwork, except as provided for under clause 2.1.4 of NZS 3604.

7. **PRIMING:**
All exterior finishing timber, all timbers in contact with concrete blockwork and all external faces, rebate, etc. of all doors, windows, frames and all woodwork of sashes, shall be primed before fixing unless otherwise specified in Painter.

8. **CLEANING:**
The Contractor at the conclusion of the Contract shall have all ceilings, walls and woodwork carefully dusted and wiped down. Windows washed and glass free from scratches and paint. Floors brushed and the entire building and site left in a clean condition for occupation.

9. **FASTENING AND FABRICATION:**
(a) The Contractor should especially note that all aspects of fastening and fabrication of timber framing members and wood-based products on this Contract shall accord with the following clauses of NZS 3604:-
Clause 2.5.1 - General
Clause 2.5.2 - Adhesives for timber & wood-based products
Clause 2.5.3 - AND APPENDIX A - for nailing and use of wire dogs
Clause 2.5.4 - for bolting and screwing.

CONTINUED...

(b) Reference shall also be made to Appendix A of NZS 3604 for the proper fixing of sheet lining and cladding materials for walls and ceilings that are not wood-based, eg: Gibraltar board, fibrous plaster or wood cellulose sheeting; especially where such materials are used as diaphragms and for wall bracing.

(c) Mild steel structural components used in sub-floor spaces, exposed to the whether or in position where condensation or dampness will occur shall be hot dipped galvanised after forming and shall provide the necessary "capacities" called for by NZS 3604 dependent on function and location.

(d) In all other cases, select and use connectors according to manufacturer's literature conforming to NZS 3604 requirements.

10. INSULATION:

All roofs, walls and floors are to be insulated in accordance with NZS 1900 Ch.4, NZS 4214 and NZS 4218 P. No insulation material shall be used that does not comply with NZS 4222. All insulation materials are to be installed in accordance with manufacturer's recommendations.

11. MOISTURE CONTROL:

Where a vapour barrier is required in the form of foil or polythene film, this barrier shall be fixed according to manufacturer's instructions. Foil or polythene film shall not be used as a substitute for building paper. Building paper shall be properly fixed to bottom plates, especially in veneer construction, to prevent entry of sub-floor air into stud cavities. Brick veneer cavities shall be closed off at and by means of soffits so as not to connect with roof cavities. There should be no unsealed openings for service pipes and wiring, etc. There should be no discharge from cooking extractors or clothes dryers into any wall, roof or floor cavity. Linings shall not be fixed until moisture content of framing has been tested and approved.
A proper solvent based paint or varnish finish must be used for plasterboard (Gibraltar board or fibrous plaster) linings to 'wet' rooms such as kitchens, bathrooms and laundry.

Water based emulsion paints must not be used.

IMPORTANT NOTE: A very valuable book entitled "The Healthy House Book", a guide to ventilation, heat and insulation in New Zealand homes can be purchased from the Building Centre.

TIMBER SCHEDULE COMMENCES PAGE NO.12

SCHEDULE OF TIMBERS (For Guides Only) LOCATION	mm x mm	GRADE AND TREATMENT For alternatives refer to NZS 3602	REMARKS
SUBFLOOR FRAMING		ALL subfloor framing No. 1 Radiata Pine* Pressure Treated to Commodity	
Wall Plates	100 x 50	Specification H3 of T.P.A. Specification	Bolted or dowelled
	min 75 x 50	e.g. Tanalith or similar	For 75mm wide studs only.
Bearers surpporting jackstuds	100 x100		
on piles	min 100 x 75		Can be 2/100 x 50's
Subfloor jackstuds	min 100 x 75		
Stud in subfloor wall	min 100 x 50		At 600 c/cs
Ditto where doubled under bearer	2/100 x 50		
Top plate to jackstuds or subfloor wall	100 x 50		
Bearers on jackstuds or timber stud subfloor wall	100 x100		
Stringer to side of foundation wall	125 x 50		Fixed with M12 bolts at 1M spacing for single story only.
Subfloor braces	min 100 x 75		Not exceeding 3M Fixed with M12 bolts (or appropriate nail plates).
GROUND FLOOR FRAMING			
Floor joists for max.1.5 KPa loading	150 x 50)	Treated No. 1 Radiata Pine to H1 of T.P.A. Specification e.g. Tanalith or Boron Type treatments.	600 c/cs. Up to 2.6m Max. span where continuous over 2 or more spans.See plan for other sizes and spans. (Along subfloor bracing lines, at right angles to joists and mid point or any span exceeding 2.5M and under load bearing walls).
Herringbone strutting	40 x 40)		
Solid Blocking	150 x 50)		
Boundary joists	150 x 25		At outer ends of floor joists.
FLOOR DECKING Refer to plans for type and location.			
a) T&G strip	100 x 22	Dressing A. Heart Rimu or Matai	Dry dressed
B) Particle Board	2400 x 1200 x 20	High Density	Joins between sheets made over supports or on 50 x 50mm blocking between joists.
WALL FRAMING			
Bottom Plates	100 x 50	No. 1 treated Radiata Pine	Long lengths
	min 75 x 50		

CONTINUED..

0002 0535 0052 2000

LOCATION	SIZE mm x mm	GRADE AND TREATMENT For alternatives refer to NZS 3602	REMARKS
WALL FRAMING - continued			
Studs	100 x 50	No. 1 or No. 2 or Building Radiata Pine	
	75 x 50	as appropriate Treated	600 c/cs
Top plates	100 x 50	PLUS 150 x 40 All No. 1 Treated	Where supported upper floors, heavy roofs
	100 x 50	Radiata Pine	and where ceiling battens necessary
	75 x 50		
Trimming studs*	100 x 50	Treated No. 1 Radiata Pine	Up to max 3m span of lintel in single storey, top storey or non load brg. wall only.
*See Remarks column for greater spans or in other locations refer to Table 15 of NZS 3604 and use doubling studs as necessary.			
Lintels (generally)	2/100 x 50	Treated No. 1 Radiata Pine	Select built-up lintels from Table 16 NZS 3604
dependent on span	up to 2/200 x 50		
Sill and Head		Treated No. 1 Radiata Pine	
Trimmers	same span as studs, thickness per span as follows;		Head trimmers shall be used instead of lintels only in non load bearing walls.
Span 2M.....40mm			
2.4M.....50			
3M.....75			
3.6M.....100 or (2/50mm)			
Dwangs (or noggings)	50 x 50	Treated No. 1 Radiata Pine	Min 2 rows for 2.4 high studs except where
	75 x 40		a greater or lesser number indicated by lining or cladding manufacturer.
Bracing	100 x 25	Treated No. 1 Radiata Pine	(Continuous let-in brace)
(No dog legs permitted)	75 x 40	Treated No. 1 Radiata Pine	(Cut between braces in diagonally opposing pairs). (Delete both above if using continuous steel strip or steel angle).

ROOF FRAMING: To pitch and style shown on plan Note 1) individual requirements and options permissible under NZS 3604 preclude meaningful scheduling here. Designer and Builder shall refer to NZS 3604 and dimension drawings to satisfaction of Territorial Building Authority. 2) All timbers as relevant in (a) (b) or (c) below shall be treated No. 1 Radiata Pine or shall be Douglas Fir. Exposed ends of Douglas Fir rafters in Type c roofs shall be capped.

(a) Trusses - To approved size and dimensions as shown on plans and manufactured by.....

or

(b) Framed (couple-close) Roofs

or

(c) Monopitch, skillion and exposed rafter roofs) as fully detailed on drawings in accord with NZS 3604. NOTE. Purlins (roof battens) on edge or on flat shall be sufficient to accomodate the thickness of thermal insulation material necessary to comply with by-laws, advise and agree with owner.

CONTINUED...

LOCATION	mm x mm	GRADE AND TREATMENT For alternatives refer to NZS 3602	REMARKS
ROOF BRACES			
Roof space braces	min 100 x 50)	Treated No. 1 Radiata Pine	(Alternative - specify opposing pairs of continuous steel strip braces instead of 100 x 25 roof plane braces).
Roof plane braces	100 x 25)		
Ceiling plane braces	100 x 50)		
ROOF SARKING			
Hit and Miss diagonal sarking	75 x 25	Treated No. 1 Radiata Pine or Douglas Fir.	60 degrees to ridge. If used delete roof plane braces above.
Sheet Sarking (e.g. 6mm plywood or 6mm particle board)			Again, omit roof plane braces
ROOF TILE BATTENS (or purlins)	75 x 50	Treated No. 1 Radiata Pine or Douglas Fir.	Light roof, on flat at 900 c/cs.
	75 x 50		Heavy Roof, on edge at 900 c/cs.
Dragon Ties	100 x 40	Treated No. 1 Radiata Pine	As required.
Eaves Framing	75 x 40	Treated No. 1 Radiata Pine	
Barge & Eaves Boards	150 x 25	Treated No. 1 Radiata Pine	
	up to 225 x 25		
Ceiling Battens	75 x 40		or as required by manufacturer of ceiling lining.
WEATHERBOARD: Dressin A Heart Rimu or Treated A Rimu or Matai or Imported Cedar. Horizontal or Vertical Finishing Grade. Weatherside or Hardiplanks. (Delete that which is not applicable).			
INTERNAL DOORS			
Jambs	25mm	(Dressing A Rimu	
Architraves	75 x 12mm	(	
Skirtings	75 x 12mm	(or by agreement	
Sill Boards	25mm	(	
Aprons	25mm	(	
Window Jambs	150 x 40mm	D.A.H. Rimu	
Window Sills	150 x 65mm	D.A.H. Rimu or Matai or Cedar	
Mullions	75 x 65mm	D.A.H. Rimu or Matai or Cedar	
Cornices	40mm	D.A.H. Rimu or Matai or Cedar	
Facings ext.	75 x 25mm	D.A.H. Rimu or Matai or Cedar	
Door Sills	150 or 200 x 65mm with 12mm Steel Weather Bar.		
Scribers	50 x 12mm	Totara or Heart Rimu	

12. SUB-FLOOR FRAMING:
    (a) Driven Timber Piles shall be included in this section of the work and all driving, testing, sizes and spacing of piles shall accord with Appendices C & D of NZS 3604 and drawings.
    (b) Bearers shall be 100mm x 100mm, supported at 1.3m centres along length and spacing between bearers to be max. 2.5m except where spacing and size of joists and bearers are shown differently in plan. Jack studs to be securely fixed to foundation piles and wall plates or bearers as shown. Brace with 100mm x 75mm as required. Floor Joists shall be gauged over plates to uniform levels. Joists shall be 150mm x 50mm. Laps not less than 150mm spiked. Nail plates may be used in place of lapping. Provide double joists under all load bearing walls.
    (c) Plates shall be 100mm x 50mm dowelled or bolted to foundation walls.

13. GENERAL FRAMING:
All framing to be 100mm x 50mm stud or 75mm x 50mm stud with 50mm wide studs o all openings not exceeding 3m span where supporting only one storey. Top and bottom plates to be same size as studs in long lengths, halved or nail plated at wall junctions and jointed over studs. All studs at 600mm c.c. Provide two (2) rows of dwangs to full height each wall unless otherwise stated on drawings.

14. WALL BRACING:
Provide all wall bracing as shown on drawings and wall bracing calculation sheets as is intended to satisfy the Territorial Building Authority. Keep strictly to the "Type", Lengths and Locations shown on plan and elevations.

15. WALL LININGS:
Generally shall be of 9.5mm Gibraltar Board and shall be fixed to external walls only after placement of thermal insulation (bulk insulants). (Specify if different, ie: if using Winstone Ltd Specific Thermal Design - with Gib-foil only in walls and higher levels of insulation on battened ceiling on trusses). All nailed or fixed as per Appendix A of NZS 3604 or manufacturer's instructions and stopped to manufacturer's instructions; in particular where contributing to wall bracing.

16. ROOF CONSTRUCTION: (See Drawings and Schedule of Timbers for types)
    (a) Trussed Roofs: Drawings showing clearly the type, pitch, span, spacing and overhangs of roof trusses and details of roof cladding shall be provided to the truss manufacturer .....to allow him to comply with clause 10.2.3 of NZS 3604. Thereafter, the Contractor shall match construction with the drawings and details provided by the truss manufacturer throughout all stages of fixing and bracing. The Contractor shall especially accord with the manufacturer's instructions for tying down where overhangs exceed 750mm. In all cases anchorage of all trusses to plates shall be with not less than 2/100mm skew nails plus 2/4.9mm wire dogs.
    (b) Framed Roofs: Roof to be framed up to pitch indicated on plans and to detail properly checked, bird mouthed and well fixed. Fix valley board, ridge etc. Fix both underpurlins and collar ties where shown and where rafters exceed 10 degrees pitch. Collar ties are to be at 1.8m centres or to every 3rd set of rafters whichever is the closer.
    Ceiling joists to be generally 100mm x 50mm spaced as required (refer to Plan) and to be well spiked to all plates. Provide adequate ceiling runners to spans of 2400mm and over.
    (c) Monopitch, skillion and exposed rafter roofs: Construct as drawings in accord with NZS 3604.

17. PURLINS, EAVES, GABLE ENDS:
Size of framing, spacing, overhangs and sheathing type as shown on plan.

CONTINUED...

18. FASCIA BOARDS, ETC:
Fix fascia, barge, frieze, etc. to suit roofing selected (See ROOFING CONTRACTOR).

19. POST AND BEAMS:
Where within maximum permitted by NZS 3604 secure in equivalent manner and with equivalent materials as in Clauses 7.1, 7.2 and 7.3 thereof. Owner to provide "specific design" for all other posts and beams and Contractor to fix as per details given.

20. FLOORING:
Timber joisted ground floors shall be insulated with ..... (brand) type perforated aluminium foil carefully laid, lapped and drooped 100mm between joists and stapled on top side, all per manufacturer's instructions before laying flooring.

(a) Particle Board Flooring shall be 20mm High Density. All aspects of handling, storage, sheet layout and fixing shall be to manufacturer's recommendations.
Nailing or stapling of pre-laid particle board floorings shall be done in two (2) stages to reduce the risk of floors squeaking. First secure at perimeter of floor, along the bottom plate positions for internal walls and sufficient elsewhere to hold the sheets flat during construction. Complete remainder of fixing to manufacturer's specification after the house has been totally "closed-in" and just prior to sanding. The Contractor shall be responsible for any damage done and for cost of repair or replacement of the flooring if not closed-in sufficiently early (normally within 2 months of laying), or if the floor is not otherwise adequately protected up to completion of the house.
Heads of fixings shall be punched to allow a reasonable depth of stopping, unless otherwise stated.
Ceilings below particle board (or any other flooring) shall not be secured until all nailing or stapling above has been done.
OWNER TO SPECIFY ANY ALTERNATIVES TO 20 (a) eg:-
(i) Particle board, laid only after closing in
(ii) Tongue & Grooved (T & G) flooring, with or without secret nailing
(iii) Glue laminated timber flooring
(iv) As (ii) and (iii) above, but also forming ceiling to a room or number of rooms below.

(b) Finishing of Particle Board Flooring
Contractor to follow manufacturer's recommendations and sequence in sanding, sealing, stopping and application of coatings to provide a minimum 3 coat flat, smooth hard wearing polyurethane finish. Contractor should also follow manufacturer's recommendations for floor areas to be covered with vinyl flooring or carpeting as advised by Owner and such finish of the particle board to be approved before either is laid.

OWNER TO AMEND 20 (b) if wishing to undertake all or part of finishing personally.

21. TIMBER TERRACES:
If shown on plan, construct to detail or best trade practice. All nail fixing with galvanised nails, all bolt fixing with galvanised bolts, all timber to be treated to the correct preservative retention as specified by the Timber Preservation Authority.

22. EXTERIOR WALL COVERINGS:
(a) Breather type building paper and masonry veneer shown on drawings shall be as given under "BRICKLAYER".
NOTE 1. Other exterior wall covering shown on the drawings and mentioned in Section 8 of the NZS 3604 should be fixed in accordance with all requirements therefore. It is important to appreciate that reference is made to NZS 3602 wherein advice is given on width of timber weatherboard and finishing. Should the Owner specify overwide boards and any finish other than paint and not adequately maintain, the responsibility for any resultant poor performance should not be apportioned to the "Contractor".

CONTINUED...

NOTE 2. Reference is made to wood cellulose flat sheets in NZS 3604 and should be taken to include such sheets that can be purchased with various finishes already applied, eg: aggregate finish or imitation stone work and brick work finishes, as well as where sheets are used as a base for insitu finishes.

(b) Other exterior wall coverings shown on the drawings shall be.....
fixed in accord with .....
and finished with .....

23. **EXTERIOR JOINERY:**
All windows shall be to NZS 4211 to suit location.

(a) Aluminium Joinery
Provide P.C. sum of $..... for windows and glazed doors to size, type and location shown on plan. Obtain joinery from .....,
complete with glazing, frictional restraints and catches, and fix plumb both ways and in correct alignment and flash all in accord with manufacturer's instructions. Sliding patio doors shall be safety glazed and fixed to timber sub-frames as per Code. NOTE: Owner should check with manufacturer regarding manner or type of safety glazing (also in shower screens). Also seek advice on satisfying the need for simple secure means of maintaining a little ventilation at night or when house is empty. Options can include substituting double spur catches for single, fixing security fasteners or provision of small sliding planes, or louvre or grill at high level.

(b) Timber Joinery
Unless otherwise specified sashes and frames shall be of finger-jointed radiata pine treated to the requirements of NZS MP3640, primed before leaving the factory, and to sizes and type shown on plan. Fix window frames plumb both ways to wall framing at not more than 750mm intervals with full width folding wedges, or to heart Totara dovetailed grounds in masonry walls. Cut all facings and scribers to fit neatly and fix quadrant under external sills. Provide and fix standard galvanized steel flashings unless otherwise specified.
Hang all sashes as required with suitable hinges or frictional restraints and fix stays and catches as supplied under P.C. sum (see Hardware).

24. **DOORS AND FRAMES (TIMBER):**
External door frames shall be of 40mm full width material rebate and internal shall be 30mm rebated or 25mm (or 18mm) material with 12mm planted stops. Doors shall comply with NZSS 1158, where external shall be framed, ledged and braced or timber or glass panel as shown on drawings, all properly constructed with stiles and rails out of 50mm material. Internal doors shall not be under 30mm hollow-core faced with plywood or hardboard with clashing strips fixed to lock stiles. All timber core material shall be treated and doors shall be of approved manufacture. Doors over 1400mm in height shall be hung on one and one half pairs of 90mm x 30mm antique butt hinges.

25. **CEILINGS:**
(a) Gibraltar Board: Fix painting quality 9.5mm Gibraltar board to joists and rafters with 30mm galvanised gib clouts, with nails punched and stopped and all joints flushed up to a true even smooth surface. In particular fix all ceilings serving as structural ceiling diaphragms under terms of NZS 3604. (See Winstone Wallboards Ltd taped joint system). Check that each wall under or connected to has adequate bracing for a diaphragm.

(b) Pinex: First quality tiles or planks as directed by Owner to be fixed to manufacturer's specification. In particular note where dragon ties have been specified on drawings and refer to NZS 3604 for their proper fixing. Also check that each wall relative to the dragon ties have the prescribed bracing values.

(c) Fibrous plaster: (See Fibrous Plaster Section).

26. **MANHOLE:**
Provide manhole in ceiling 500mm x 500mm where directed.

CONTINUED...

* 27. **WARDROBES:**
To be lined full height. Provide inside each with 300mm x 25mm full width shelf at 1.7m from floor and 20mm galvanised pipe coat rail at 75mm below shelf. Provide cupboards over wardrobe where required.

* 28. **KITCHEN CUPBOARDS:**
Construct cupboards and pantry unit to sizes shown on plans, with doors and shelves to Owner's approval.

29. **SINK TOP:**
Provide for standard sink top. Unit to have standard 450mm x 300mm stainless steel sink. Fix on unit either indicated on plan or to Owner's final approval. Units to be generally 900mm high and 500mm in depth, 100mm toe space. Provide flush cupboard doors to front and shelving to approval. Timber used in doors to be approved by Owner.

* 30. **HOT WATER CUPBOARD:**
To be constructed where shown and fitted with slat shelving above the cylinder spaced at 15mm apart. Provide two flush doors with thermostat boxed in.

31. **ARCHITRAVES, SKIRTINGS, ETC:**
Finish all windows internally, door openings and wherever required with 40mm x 12mm rounded or splayed architraves, or as otherwise specified..... Finish at junction of floor and wall with 60mm x 12mm or 40mm x 12mm, with ends neatly butted, scribed or mitred. Scribe to floor where necessary. Where required fix scotia mould at ceiling junction, sup 'fix beads, half rounds and all trim as required to complete the work.

* 32. **BATHROOM; ENSUITE; SHOWER AREAS:**
Provide and fix in bathroom one toilet cabinet 550mm x 400mm recessed into wall with mirrored door. Bath to be built in and supported with 75mm x 50mm framing on edge. Fix bath panels as agreed by Owner. Allow for toe space. All wall linings to be fixed to manufacturer's specification with adequate capillary gaps or flashings/trim or sealant to prevent leakage. Provide for all sanitary fittings shown on plan and given under "Plumber".

* 33. **LAUNDRY AND TOILET/S:**
Fit out as shown on plan and given under "Plumber".

34. **METER RECESS:**
Provide recess for Electric Meter Board where directed to the satisfaction of the Local Electric Supply Authority.

35. **HARDWARE:**
Allow the P.C. sum of $..... for the supply of all hardware required. Door handles, locks, window fittings, door and drawer pulls, stops, angles, vents, towel rails, etc. Allow to take delivery and fix. The supply of hinges and butts for all doors and windows is not included in the P.C. sum.

36. **INTERNAL WOOD STAIRS:**
Consult with Owner on construction of internal wooden stairs.

* **IMPORTANT NOTE:**
For Clauses 27, 28, 30, 32 and 33. Allow for door of full height and full width access to wardrobes, pantries, linen cupboards, etc. Then consider provision of adjustable shelving, draw and basket fittings as budget allows. This will minimise further alteration.

ROOFING CONTRACTOR

1. PRELIMINARY AND GENERAL:
Read and note all clauses under Preliminary and General of this specification where they apply to this trade. (Also see Plumber).

2. ROOF FRAMING AND WALLS:
Roof framing shall provide adequate support and fixings for purlins (roofing battens). No member shall be overloaded by landing heavy, localised, stacks of cladding on them prior to fixing, and temporary braces as might be necessary, shall be provided in walls below to sustain such loading until roof cladding is complete and wall lining subsequently fixed.
In roof types (B) and (C) of timber schedule, the size of purlins if not shown or stated shall be such as to accommodate the required thickness of thermal insulation to comply with the bylaws; advise and agree with Owner.

3. ROOFING UNDERLAY:
Underlay shall be breather type building paper to NZS 2295. Run horizontally with upper sheets lapped 75mm over lower sheets and with bottom edges turned over fascia into gutters. Such underlay shall be provided under all metal roofs and be adequately supported unless self-supporting type.
Where other roof claddings are specified and where at less than 12.5 degree pitch undertake all work in accord with a method approved in writing from the cladding manufacturer for the particular job.

4. PREPARATORY WORK:
Provide all gutters, valleys and underflashings before cladding commences. Except where stated, valleys shall be of 0.6mm galvanised steel at least 400mm wide with folded edges and with lower end finishing well into spouting. Use 0.7mm aluminium valley for aluminium roofing. For profiled metal roofing set out purlins to give reduced span at eaves and between top purlin and ridge as recommended. For tiled roofs provide anti-ponding boards at eaves as shown to support building paper. Allow tiles to adequately project into spouting and set out so that when laid there is full height course of tiles both at eaves and ridge. Ensure all edges of roof cladding are adequately supported around projections such as pipes, ducting and roof lights.

5. SARKING TO ROOFS, DECK AND GUTTERS:
Profiled metal roofing shall not be laid at less than 5 degree pitch except where shown and only when adequately supported by hit-and-miss type sarking or full sheet sarking so as to prevent ponding.
Sarking for roofs, decks and gutters that are to be covered by Nuralite or Butynol shall be of material and fixed in a manner approved by the manufacturer.

6. ROOF CLADDING MATERIALS AND COMPONENTS:
Roof cladding shall be ..... (nominated colour and type), manufactured by .....
Materials selected shall be set out and fixed by approved FIXERS and shall accord with the manufacturer's specification and in all cases with NZS Specifications or others where appropriate, ie:-

NZS 3403: 1978 Hot-dipped galvanised corrugated steel sheets for building purposes
NZS 3441: 1978 Hot-dipped zinc-coil steel coil and cut lengths
NZS 4206: 1973 Concrete interlocking roofing tiles
NZS 4217: 1980 Pressed Metal Tile Roofs
BS 4300/6: 1969 NS31 sheet and strip (Part 6 is one of a series covering wrought aluminium and aluminium alloys).

CONTINUED...

Thickness of material unless otherwise specified shall not be less than the following:-

Aluminium cladding, valleys and flashings	0.7mm
Galvanised steel cladding, spouting, downpipes and overflows	0.6mm
Copper flashings, gutters and sumps	0.6mm
Lead flashings to small diameter vent pipe with PVC flashing cone	1.7mm
Other lead flashings and soakers	2mm

7. UNPAINTED GALVANISED STEEL ROOFING:
Areas which are inaccessible for maintenance such as laps in the roofing cladding and between cladding and metal flashings and cappings shall be primed and top coated on both surfaces before fixing.

8. FACTORY PREFINISHED MATERIAL:
Treat all such material with great care and obtain matching colour for making good minor damage and covering fixings made through the face of the claddings.

9. CAPPINGS AND OVERFLASHINGS (ALSO SEE PLUMBER):
Adequately and neatly secure all ridgings, cappings and overflashings wherever needed to make and keep roof watertight. As much as possible use only material compatible with the roof cladding. If dissimilar metals are used keep from making contact.
Sealants shall not be used except where approved. Remove all traces of flux after welding, soldering or brazing. Do not leave swarf, broken rivets, screws, nails or waste metal on roof, nor in gutters or spouting.

10. BIRD PROOFING:
Where indicated bird proof eaves with purpose-made foam plastic impregnated with bitumen.

11. SPOUTING AND DOWNPIPES - SEE PLUMBER

12. GUARANTEES:
On completion of all roofing thoroughly clean down and furnish Owner with written guarantees to watertightness and security of the roof cladding. Guarantees to be signed by the approved FIXER/FIXERS and run for a period of three (3) years from the date of completion of the building contract.

PLUMBER

1. PRELIMINARY AND GENERAL:
Read and note all clauses in the Preliminary and General of this specification which shall apply to all work in this section. (ALSO SEE ROOFING CONTRACTOR)

2. GENERAL:
Inclusion of roof flashings, provision of roof drainage and other general flashing requirements in this section shall mean that all shall be of the quality expected of work customarily done by plumbers. The foregoing shall apply whether or not the ROOFING CONTRACTOR employs a plumber to execute or supervise, or if the builder himself fixes flashings to windows and external doors. All other work shall conform to the requirements of the Territorial Authority under Model General Bylaw NZS 9201: Chapter 7: "Water Supply" and Drainage and Plumbing Regulations 1978.

▶▶ No connections to sanitary fittings or main shall be made except by a Registered Plumber. Plumber shall arrange all permits, connection and testing fees.

3. FLASHINGS:
Wherever possible use flashings that are readily available but purpose-made by the roof cladding manufacturer to suit the roofing material and profile selected. Otherwise provide flashings as recommended but not supplied by cladding manufacturer and adequately secure. Flash wherever needed to make and keep roof watertight. Prime and topcoat both surfaces at laps with galvanised steel.
Flashings to each window head and door shall be provided in one piece, each 150mm longer than the head it is to protect.

4. SPOUTING AND DOWNPIPES:
Spouting shall be that manufactured by the roof cladding manufacturer, or standard 125mm quadrant type galvanised steel, or approved white PVC. Provide and fix as recommended with even fall to downpipes. Provide all necessary stopped ends, mitred returns and outlets and ensure all joints in metal spouting are adequately soldered or welded, or for PVC are made with solvent cement except where joints provide for thermal movement. Connect outlets to 80mm diameter downpipes. Avoid high local discharge from high roof to low roof by use of a spreader pipe. Provide sufficient number of downpipes to service roof areas.

▶▶ Discharge stormwater into soakaways or street channel, NOT into foulwater drainage system UNLESS permitted by Territorial Authority.

NOTES AND ASSUMPTIONS RELEVANT TO FOLLOWING CLAUSES 5 - 10

IMPORTANT

Very careful consideration needs to be given to the design of household plumbing and selection from a wide range of materials nowadays available in supply pipe for both hot and cold water services; and from a similarly wide range of materials for sanitary fittings and waste pipes from them. Every effort should be made to consult with manufacturers and plumbers BEFORE completing this specification in order to attain plumbing to suit individual circumstances. Territorial Authority plumbing inspectors can advise on such matters as water pressure and volume obtainable from mains and materials approved. The following is offered for guidance to help in the completion of the plumbing specification similar to the clauses given but other clauses might also be necessary if choice differs from the assumptions made.

(a) Cold water can be obtained from a Territorial Authority mains normally providing adequate water pressure and volume; so avoiding the need for a cold water supply tank in or on the roof.

▶▶ (b) Water heating is to be mainly by means of a low-prescure thermal storage electric water heater; with cold water feed into Hot Water Cylinder through a pressure-reducing valve and separate non-return valve. Alternatives include:-

CONTINUED...

B1. Use of a break-pressure tank installed above the Hot Water Cylinder OR
B2. A mains pressure water heater; with check valve (gate valve) between the two (2) stop valves OR
B3. Provision for gas water heater or heaters OR
B4. Provision for a wet-back, eg: from free-standing fireplace or stove, or solar water heating; supplementary to main method of water heating OR
B5. Water heating for space (room) heating as well as serving sanitar, ittings.

(c) Proximity of sanitary fittings to Hot Water Cylinder will normally be in order of most frequent use, eg: kitchen sink, bath or lavatory basin, bath/shower, tub and washing machine.

(d) Cold and hot water supply to ALL fittings is assumed to be by 12mm nominal bore pipes with temperature-compensating mixer valve to shower to reduce severe temperature changes when taps are turned on or off elsewhere in house. This practice is now common in many areas having reasonable water pressure. Check if permitted and satisfactory in your area and discuss alternative methods of reducing shower temperature fluctuations (by pipe sizing and positioning) with plumber.

5. COLD WATER SUPPLY:
Tap off from mains service pipe in ..... mm COPPER/POLYBUTYLENE/UPVC/or TYPE 3 or TYPE 5 POLYETHYLENE (Polythene). Run pipe not less than 450mm below finished ground level to the house with entry through or under the wall foundation. Supply and fix a water meter, box and cover when required by Territorial Authority.
Provide ..... mm ..... pipe to hose taps at front and rear of building and to all fittings including hot water cylinder and washing machine. All branches to be as short, straight and at as even gradient as possible with easybends used throughout (NOT ELBOW FITTINGS). Use only approved connections throughout. All pipes are to be adequately supported, well secured and where possible concealed.
When using any form of plastic piping, plumber to discuss and provide means of ELECTRICAL EARTHING with electrician.

6. HOT WATER SUPPLY:
Supply one (1) pressure electrical hot-water cylinder of ..... litre capacity of approved make and type and install and connect all pipes to the approval of the Territorial Authority. Provide ..... mm COPPER or POLYBUTYLENE hot water supply pipes and branches to all fittings including washing machine.

7. SANITARY FITTINGS, ETC:
Provide P.C. sum of $..... for the following. Carefully note dimensions and install where shown on plan and to Owner's approval. (Specification should preferably include the number of each required size, brand, colour, material, and where necessary height).

..... Bath/s mm x mm ....
Vitreous China/Porcelain enamelled pressed steel/acrylic

.. .. WC/s ....
Vitreous China complete with ..... double flap plastic seat

..... WC Cistern/s ....
PVC complete with ..... PVC flush pipe or chrome plated flush pipe

..... Basin/s (on wall brackets) (on pedastal) (or in vanity)
..... mm x ..... mm
Vitreous China/Porcelain Enamel/Acrylic

..... Shower tray/s (or Shub Tub) ....
Stainless Steel/Porcelain Enamel/Acrylic/GRP

CONTINUED...

SANITARY FITTINGS (Continued)

..... Bidet ....
Vitreous China

..... Sink/s and top ..... with single or double drainer
As per plan with laminated top/stainless steel/or tiled or mosaic top. Height ..... mm

..... Laundry tub ..... mm x ..... mm single or twin tub combination.
Stainless Steel/Porcelain enamelled, on stand, with cupboard below. Height ..... mm

8. TAPS, FAUCETS AND VALVES:
Consult with Owner on choice of brands and fixing of all taps, faucets and mixing valves. Provide and fix all other water supply fittings as necessary.

9. WASTES AND VENTS:
Provide all necessary taps, waste pipes, soil stacks, back vents and terminal vents. Provide cleaning eyes to all waste pipe at junctions and all necessary overflow pipes.

10. OTHER MEANS OF HEATING WATER:
Consult with Owner on choice and fitting of free standing fireplace with wet-back/or stove with wet-back/ or solar water heating panels to supplement hot water heating system.

DRAINLAYER

1. PRELIMINARY AND GENERAL:
Read and note all clauses under Preliminary and General of this contract where they apply to this trade.

2. EXTENT OF WORK:
Work in this section of the contract comprises all surface and foul water drainage up to above ground level to connect to Plumber's work. Include all pipes and specials, fittings, construction of manholes, all gully traps and connections for terminal vents, soil and waste pipes. The Drainlayer shall confer with the Plumber and shall arrange with the Contractor before the foundations are laid to fix the exact position of all connections of wastes and drains.

3. STANDARD OF WORK:
The whole of this work shall be carried out by experienced tradesmen to the satisfaction of the Owner and the Territorial Authority. It shall conform to requirements of the Drainage and Plumbing Regulations of 1978 and the Local Council Specification, the Contractor shall allow accordingly should conflict exist. Obtain all permits, service all notices and pay all fees required and arrange for all tests.

4. MATERIALS:
All materials shall be the best of their respective kinds. All cast iron pipes shall be free from rust and be of first quality and of even wall thickness and shall be hot-dipped. Those for the use in foul drains shall have a wall thickness of not less than 4.7mm. Cast iron fittings shall be of similar quality and have inspection plates as required. General drains shall be 100mm diameter first class glazed earthenware with rubber ring joints.

CONTINUED...

FIBROUS PLASTERER

1. **PRELIMINARY AND GENERAL:**
Read and note all clauses under Preliminary and General of this specification where they apply to this trade.

2. **WORK INCLUDED:**
The manufacture and application of fibrous plaster or plasterglass sheet to wall or ceiling surfaces as specified. The manufacture and application of cornices or other decorative fibrous plaster items as specified.

3. **MATERIALS AND WORKMANSHIP:**
All fibrous plaster or plasterglass sheets and other products shall be manufactured and fixed, strictly in accordance to NZS 4221 or with the Code of Practice of the New Zealand Fibrous Plaster Manufacturer's Association. The whole of the labour required for the erection, fixing, wadding and stopping shall be that of competent fibrous plaster tradesmen. Admixtures, release agents and stopping materials shall not be of a deleterious nature nor used in quantities sufficient to impair the properties of the sheet when used with or without decoration.

4. **FRAMING:**
All noggings, trimmings, straightening and packing of studs or joists, necessary for the fixing of fibrous plaster products shall be provided and completed by the Builder before the commencement of such work. Timber framing shall comply with NZSS 3631 (framing grades) and shall be pre-dried to an equilibrium moisture content not exceeding; 15-16%. The surface to which fibrous plaster is fixed must be clean, straight and dry.

5. **CEILING DIAPHRAGMS:**
Fix all ceilings serving as structural ceiling diaphragms under terms of NZS 1604 (obtain methods of fixing and size limitations from manufacturer). Check that each wall under or connected to has bracing for a diaphragm.

6. **PAINTING:**
All fibrous plaster or plasterglass sheets and other products shall be painted strictly in accordance with the specifications outlined below:-

SYSTEM	FIRST COAT	SECOND COAT	THIRD COAT
No. 1	Varnish based Pigmented Sealer	Primer Undercoat	Full Gloss
No. 2	Varnish based Pigmented Sealer	Semi Gloss	Semi Gloss
No. 3	Varnish based Pigmented Sealer	Alkyd Flat	Alkyd Flat
No. 4	Varnish based Pigmented Sealer	PVA Plastic	PVA Plastic

Allow overnight drying between coats except for system No.4 where one or two hours drying time, according to manufacturer's instructions may be allowed between second and third coats.
NB: The fibrous plasterer will not accept responsibility for the effect of glancing light on fibrous plaster with a gloss paint finish.

ELECTRICIAN

1. **PRELIMINARY AND GENERAL:**
Read and note all clauses under Preliminary and General of this specification where they apply to this trade.

2. **FEES:**
Pay all fees and charges and obtain all necessary permits for this trade.

3. **SCOPE OF WORK:**
Carry out the whole of the electrical installations in strict accordance with the latest Electrical Wiring Regulations and Territorial Authority's By-Laws, and meter wiring diagrams.

4. **MATERIALS AND WORKMANSHIP:**
All materials used under this contract shall be of approved British or New Zealand Standard Specification. Allow for materials necessary to complete the Contract, whether specified or not. All work shall be carried out by a Registered Electrician in accordance with regulations and best trade practice and in a manner which will cause minimum inconvenience to other workmen and the work as a whole. Do all cutting away, drilling, etc., and with timber cut the minimum only away for the entry of cables.

5. **CO-OPERATION:**
Co-operate with the Building Contractor and other sub-contractors in all phases of work. Give ample notice to enable the Contractor to arrange the necessary void, chase data, etc.

6. **COMPLETION AND CONNECTION OF POWER:**
Leave work complete, pay all charges and arrange for all inspections and tests and for the connection of power to the works. It is the responsibility of the Electrical Contractor to ensure that no delay is occasioned to the job once the Contract is completed.

7. **POWER BOARD SUPPLY:**
Arrange with the Power Board, allow for and pay all fees for the connection of underground, or overhead, supply to the residence.

8. **METER BOX:**
Provide and install recessed meter box where shown on plan. Confer with Carpenter for trimming same.

9. **MAIN SWITCHBOARD:**
Provide and install in recess main switchboard complete with all necessary control and auxiliary equipment.

10. **ELECTRIC STOVE OR SIMILAR:**
    (a) Allow the P.C. sum of $..... for the supply only of the stove. Provide and fix a 30-amp flush switch for stove and sufficient cable for connection and allow for installation.
    (b) Allow the P.C. sum of $..... for the supply of extract fan in kitchen and bathroom (or Rangehood in kitchen).

11. **WATER HEATER:**
Allow for the permanent connection of the water heater to the electrical system. Provide and install 3kw element and thermostat to ..... litres hot water cylinder provided by the Plumber. Refer clause 3 of this section.

12. **POWER POINTS:**
All wall plugs shall be 230v, 10amp, 3 pin flush type. Generally install plugs 300mm above floor or 225mm above bench top. Points to washer/dryer space and refrigerator 1200mm from floor. The exact position of all power points shall be determined on the job by the Owner.

CONTINUED...

13. LIGHTS:
All lights to be first quality, plastic batten holder shall be reinforced. All roses and holders not covered by fittings shall be white.

14. LIGHT SWITCHES:
Light switches shall be 10amp, all insulated P.D.L. micro-gap type or equivalent. Where indicated fit flush type with plain bakelite flush plate. Fix switches generally 1200mm above floor.

15. SPECIAL FITTINGS:
Allow the sum of $ ..... for fittings to be selected by Owner.

16. LIGHT POINTS:
Allow for installing a total of ..... outlets in building.

17. POWER POINTS:
Allow for installing a total of ..... power points in the building.

PAINTER AND GLAZIER

1. PRELIMINARY AND GENERAL:
Note all clauses under Preliminary and General of this specification which shall apply to this section of the work.

2. MATERIALS:
Generally all materials shall be of NZ manufacture of approved brands and of the paint selected respective coats shall be of the same brand.

3. WORKMANSHIP:
All work shall be carried out in accordance with good trade practice. Surfaces shall be clean and properly prepared before coating and work shall be in accordance with Code of Practice NZS 7703 "Painting of Building".

4. COLOUR SCHEME:
The Owner will select from standard colour charts, colours he will require and Contractor is to allow for picking out sashes, doors, porches or any other reasonable colour change required.

5. STOPPING:
After priming, all nail holes or joints are to be stopped and cleaned off before undercoating for painted work and for varnished work, holes, etc., are to be stopped with matching putty after first coat of sealer.

6. PAINTING OF EXTERNAL WOODWORK:
After priming all external woodwork and adjacent metal work such as flashings, spouting and downpipes is to be given one good coat of undercoat followed by finished coat of high gloss paint. Priming coat before painting shall be well brushed in and all faces shall be covered, ends of laps and tops and sides of sashes, doors, etc.

CONTINUED..

7. PAINTING OF INTERIOR SURFACES:
Refer to plan or separate instructions to accompany this specification. Wallboard and ceilings as required to be given one coat of sealer and finished with two coats of approved paint finishing flat or semi-gloss as required. Where full gloss is required such as kitchen and bathroom, finishing coat shall be full gloss enamel.

8. VARNISHING:
Where varnishing is required such as doors, architraves and skirtings give one coat of approved P.V.A. sealer followed by two coats of clear varnish finishing egg-shell gloss and lightly sanding between coats.

9. CONCRETE:
To concrete base walls or concrete block work apply two coats of exterior quality P.V.A. paint of an approved type finishing semi-gloss.

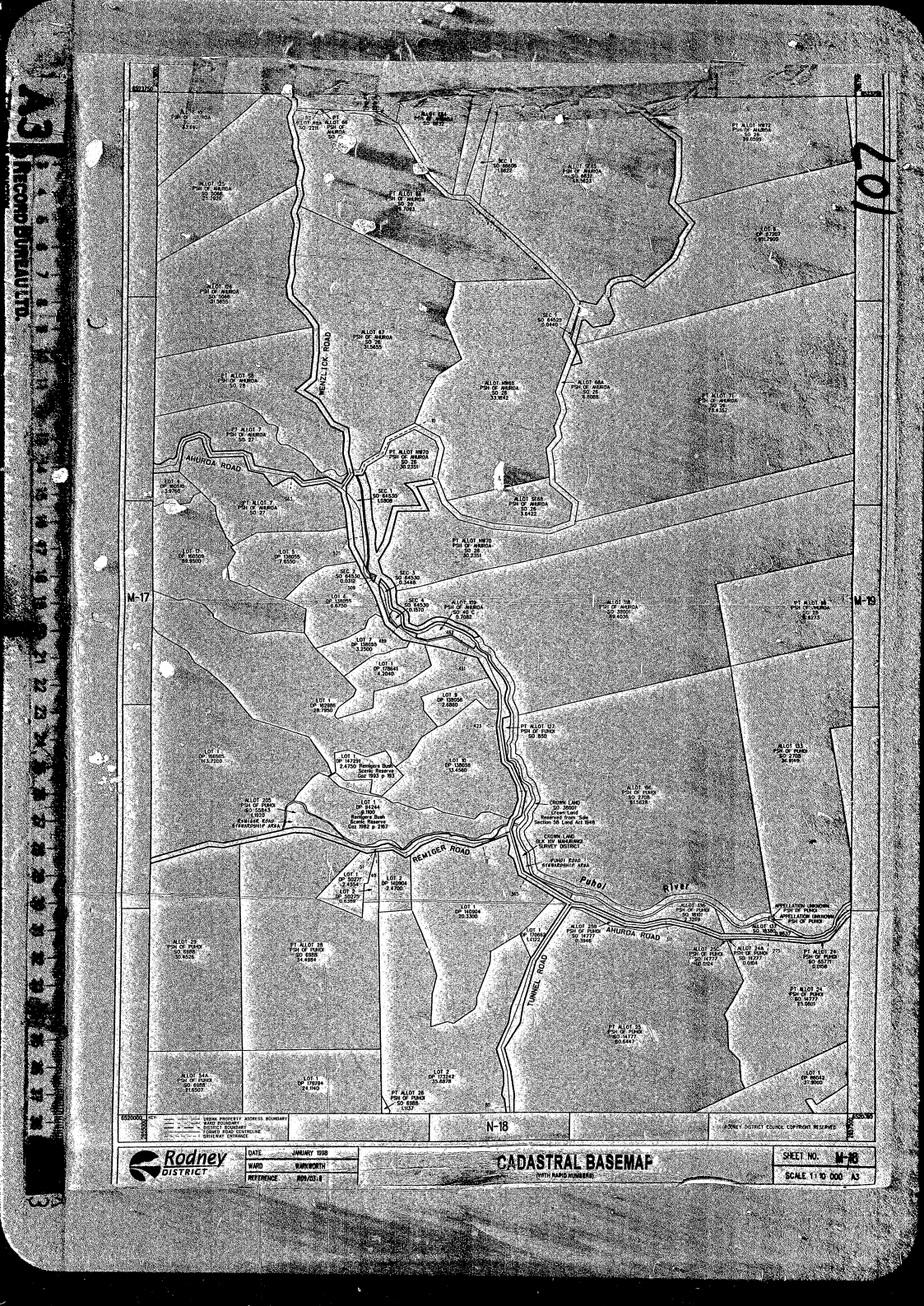
10. ROOF:
Roof - If corrugated iron prime all laps before fixing. For corrugated iron or galvanised trough section roofing after fixing prime roof with calcium plumbate galvanised iron primer and then apply one good coat of approved roof paint.

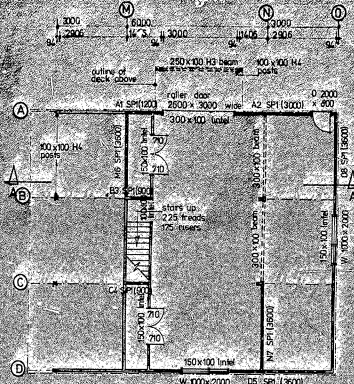
11. PAPERHANGING:
All rooms to be papered shall have paper hung to a value of $ ..... per roll. Wall shall be properly prepared by sizing, paper hung straight and true with butt joints, and paste used shall have a fungicide incorporated.

12. GLAZIER:
Glaze all window sashes, doors or screens and overhead screens, if shown, with appropriate weight glass and safety glasses or safety plastics properly fixed and puttied or beaded into rebates or gasket glaze. Where required glazing shall be selected obscure patterned glass.

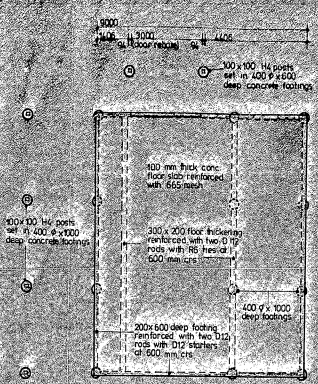
13. MIRROR:
Provide a 6mm plate glass mirror and mount on bathroom cabinet door with mirror clips.

14. COMPLETION:
The Painter and Glazier is to do all that is required of their trades to leave the work complete and all must be left clean including all glass at completion.

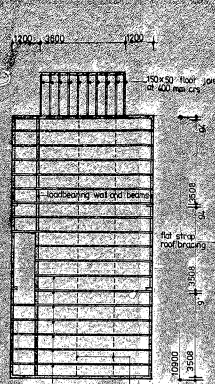

AHURORA ROAD
MEREDITH ROAD
REMIGER ROAD
TUNNEL ROAD
Pohoi River
M-17
M-18
M-19
N-18
CROWN LAND SO 38501 Reserved for Scenic Reserve Section 58 Land Act 1948
CROWN LAND SO 61 MARANGAI SURVEY DISTRICT
PUHOI ROAD SYMMONDS AREA
REMIGER ROAD SYMMONDS AREA
Remigers Bush Scenic Reserve Gaz 1983 p 193
Remigers Bush Scenic Reserve Gaz 1982 p 2767
APPELLATION UNKNOWN PART OF MARKS APPELLATION UNKNOWN PSH OF PUHOI
PSH OF AHURORA
PSH OF PUHOI
DP 17845
DP 17846
DP 17847
DP 17848
DP 17849
DP 17850
DP 17851
DP 17852
DP 17853
DP 17854
DP 17855
DP 17856
DP 17857
DP 17858
DP 17859
DP 17860
DP 17861
DP 17862
DP 17863
DP 17864
DP



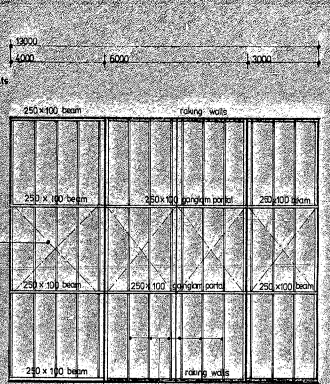
ground floor plan
scale 1:100



foundation plan
scale 1:100

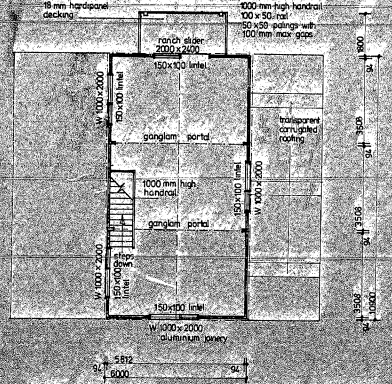


upper floor framing plan
scale 1:100



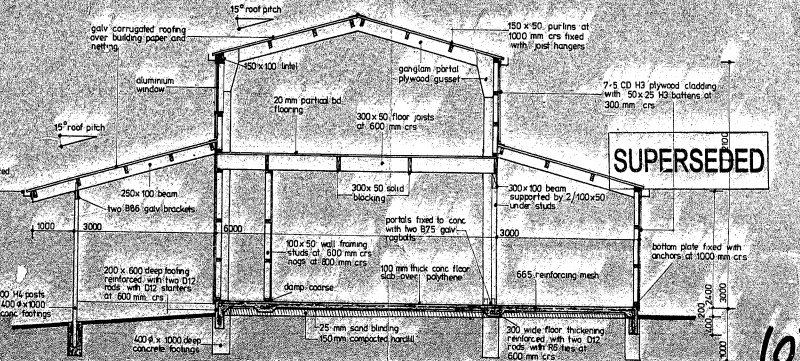
roof plan
scale 1:100

roof pitch 15°
roof material corrugated galv iron



upper floor plan
scale 1:100

GROUND FLOOR AREA 130
UPPER FLOOR AREA 65
TOTAL 195



cross section A-A
scale 1:50

SUPERSEDED

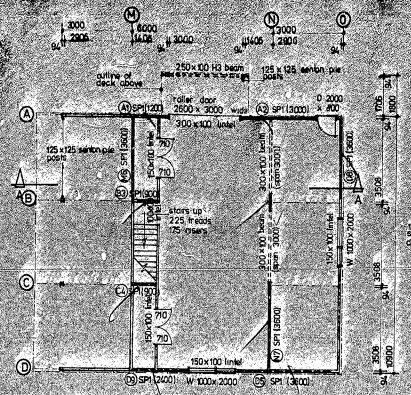
107

Puhoi Design
architectural draughting
27 Saleyards rd Puhoi
phone (09) 4220626
mobile (025) 2264475

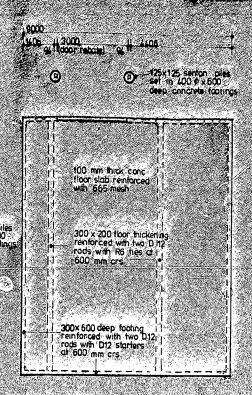
PROPOSED SHED-WORKSHOP AT REMIGER RD PUHOI FOR R NICKLIN

ground and upper floor plan foundation, subfloor, roof plan cross section	drawn D W PAPPRILL	scale 1:100, 50 ref	sheet 1
	date JULY 1998		

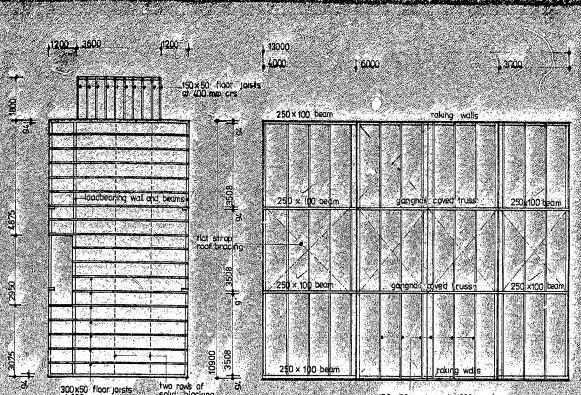
A2 MICRO RECORD BUREAU LTD.



ground floor plan
scale 1:100



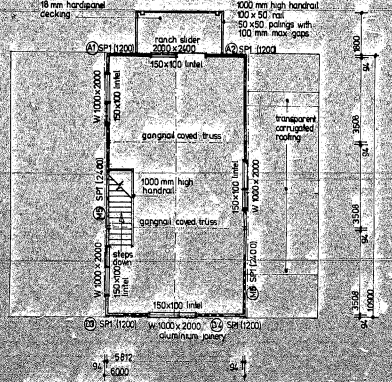
foundation plan
scale 1:100



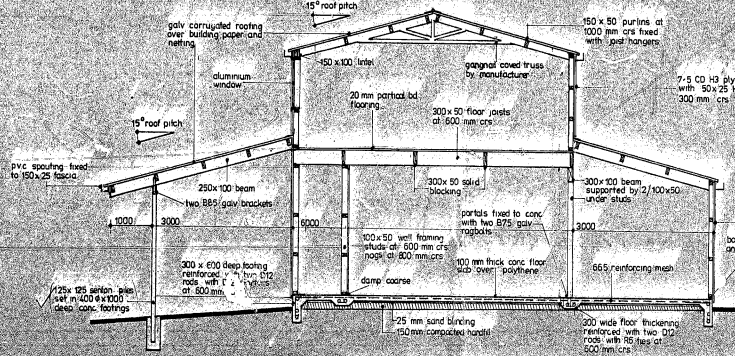
upper floor framing plan
scale 1:100

roof plan
scale 1:100

roof pitch 15°
roof material corrugated galv iron



upper floor plan
scale 1:100



cross section A-A
scale 1:50

Rodney District Council
BUILDING (ACCEPTANCE OF PLANS) ACT 1991
Approved for building consent
ABA 981674
Officer
Date 10.8.98
See Building Consent for conditions.

ground floor area	130 m²
upper floor area	65 m²
total	195 m²

Puhoi Design
architectural draughting
27 Selwyn rd Puhoi
phone (09)4229636
mobile (025)225675

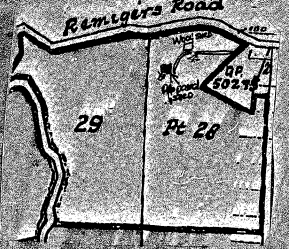
PROPOSED SHED-WORKSHOP AT REMIGER RD PUHOI FOR R NICKLIN

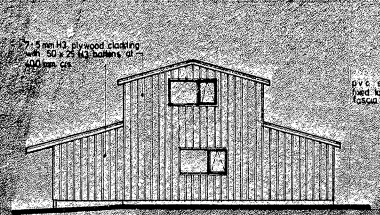
ground and upper floor plan
foundation, subfloor, roof plan
cross section

drawn D.W. PAPPRILL
scale 1:100, 50
date JULY 1998
ref
sheet 1

107

A2
MICRO RECORD BUREAU LTD.

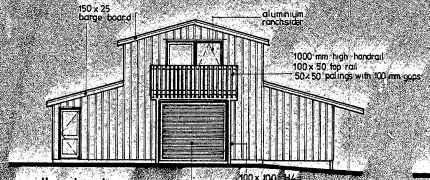




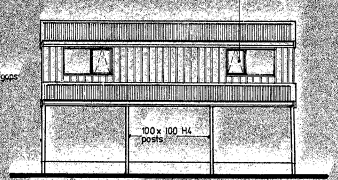
south elevation
scale 1:100



east elevation
scale 1:100



north elevation
scale 1:100



west elevation
scale 1:100

SUPERSEDED

(07)

Puhoi Design
architectural draughting
27 Selwyn rd Puhoi
phone (09)4220656
mobile (025)225475

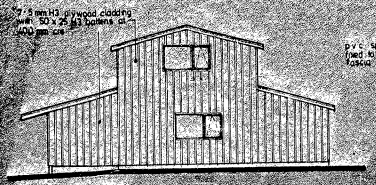
PROPOSED SHED-WORKSHOP AT REMIGER RD PUHOI FOR R NICKLIN

elevations

drawn	scale	sheet
D W PAPPILL	1:100	2
date JULY 1998	ref	

A2

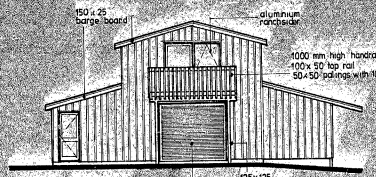
MICRO RECORD BUREAU LTD.



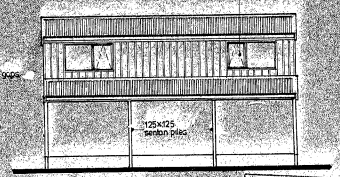
south elevation
scale 1:100



east elevation
scale 1:100



north elevation
scale 1:100



west elevation
scale 1:100

Rodney District Council
PLANS/SPECIFICATIONS
Approved for building
consent
ABA 98174
Officer BG
Date 10.8.98
See Building Consent for
conditions.

ALL RIGHTS IN THESE DRAWINGS
ARE RESERVED TO PUHOI DESIGN
AND MUST NOT BE COPIED OR
USED WITHOUT THE DESIGNER'S
WRITTEN CONSENT

Puhoi Design
architectural
draughting
27 Saleyards rd Puhoi
phone (09)4220698
mobile (025)225475

PROPOSED SHED-WORKSHOP
AT REMIGER RD PUHOI
FOR R NICKLIN

elevations	drawn	scale	sheet
	D.W.RAPPELL	1:100	2
	date	ref	
	JULY 1998		

107

MICRO RECORD BUREAU LTD.

